BIOL 220 Problem Set 05: Null Hypothesis Significance Testing

Answer Key

Identifying H_0 and H_A

Questions 1–3 will get you thinking about what distinguishes null and alternative hypotheses

1. Give an appropriate null hypothesis for this alternative hypothesis [1 point]:

   H_{A}: The mean leaf size of plants from tropical latitudes is different than the mean leaf size of plants from temperate latitudes.

Possible answer
H_0: The mean leaf size of plants form tropical latitudes is equal to the mean leaf size of plants from temperate latitudes

2. Give an appropriate alternative hypothesis for this null hypothesis [1 point]:

   H_0: There is no difference in the mean blood concentrations of C-reactive protein (used to diagnose Crohn's disease, units: mg/dL) in patients given a diet high in broccoli sprouts compared to the control group

Possible answer
H_{A}: The mean blood concentrations of C-reactive protein (used to diagnose Crohn's disease, units: mg/dL) in patients given a diet high in broccoli sprouts is different to the mean blood concentrations of C-reactive protein to the control group

3. Determine whether each hypothesis below is more appropriate as a null hypothesis or an alternative hypothesis: [1 point]

- Hypothesis: sea water temperature changes the proportion of corals that experience bleaching.
- Hypothesis: the ratio of body mass to brain mass across mammals equals 1.
- Hypothesis: the average number of animal species within tropical latitudes is the same as the average number of animal species outside of the tropics
- Hypothesis: the effect of a new experimental drug called "feel better syrup" has no effect on duration of common cold symptoms

Answer

- Hypothesis: sea water temperature changes the proportion of corals that experience bleaching. $\mathbf{H_A}$
- Hypothesis: the ratio of body mass to brain mass across mammals equals 1. $\mathbf{H_0}$
- Hypothesis: the average number of animal species within tropical latitudes is the same as the average number of animal species outside of the tropics $\mathbf{H_0}$
- Hypothesis: the effect of a new experimental drug called "feel better syrup" has no effect on duration of common cold symptoms $\mathbf{H_0}$

P-values, sample size, and whether the null is really true

Pay close attention to the details in questions 4–5!

4. Consider a clinical trial of a new drug to treat Alzheimer's. Half the participants are given the drug; half are given a placebo control. Treatments are assigned at random. Assuming that the drug actually has no effect, which of the following statements is true? [1 point]

   a. A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$
   b. A clinical trial with a larger sample is **less** likely than a smaller trial to get a result with $p < 0.05$
   c. A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$
   d. None of the above are true

Answer

c. A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$

This may seem counter-intuitive but the reason is that when there truely is no effect (i.e. when H_0 really is true), then the chance of getting $p < 0.05$ is the same across sample sizes

5. Consider a clinical trial of a new drug to treat Alzheimer's. Half the participants are given the drug; half are given a placebo control. Treatments are assigned at random. Assuming that the drug *actually does reduce the chance of developing Alzheimer's*, which of the following statements is true? [1 point]

    a. A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$
    b. A clinical trial with a larger sample is **less** likely than a smaller trial to get a result with $p < 0.05$
    c. A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$
    d. None of the above are true

Answer

a. A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$

Hawaiian *Tetragnatha* spiders

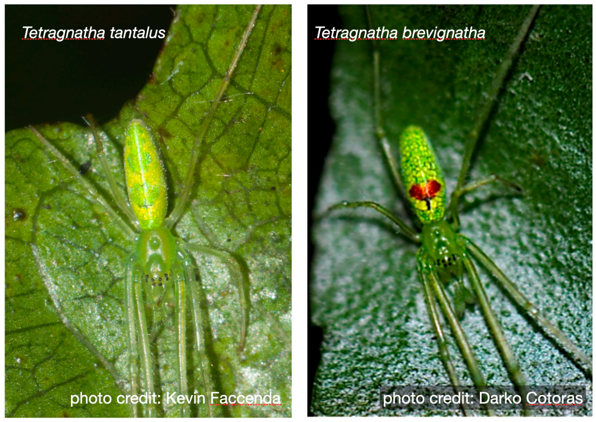


Figure 1: Two species of endemic Hawaiian *Tetragnatha* spiders

One very cool group of spiders in Hawai i are the *Tetragnatha* spiders, which are an example of adaptive radiation. On each island, these spiders re-evolved distinctive color morphs that use different strategies to avoid depredation and catch prey. The “green” color morph is especially pretty. On O ahu there are two green morphs, one is *T. tantalus* shown above. On Hawai i Island there is only one green morph–*T. brevignatha*. I used to spend a lot of time up in the uka catching and identifying spiders, but being behind the computer all the time, I worry I lost those skills. A friend gave me a challenge: they showed me 10 pictures of *Tetragnatha* and told me 5 were *T. tantalus*, and 5 were *T. brevignatha*. My test was to correctly guess which was which. Don’t be fooled! Not all *T. brevignatha* have the red spot (some are all green), and some *T. tantalus* have a red spot of their own!

Out of 10 trials, I got 7 right. My “scientific” hypothesis is that I still know how to ID spiders, but statistically, is that really supported by these data?

Use the data from this totally true, not made up, spider identification test to answer questions 6–10.

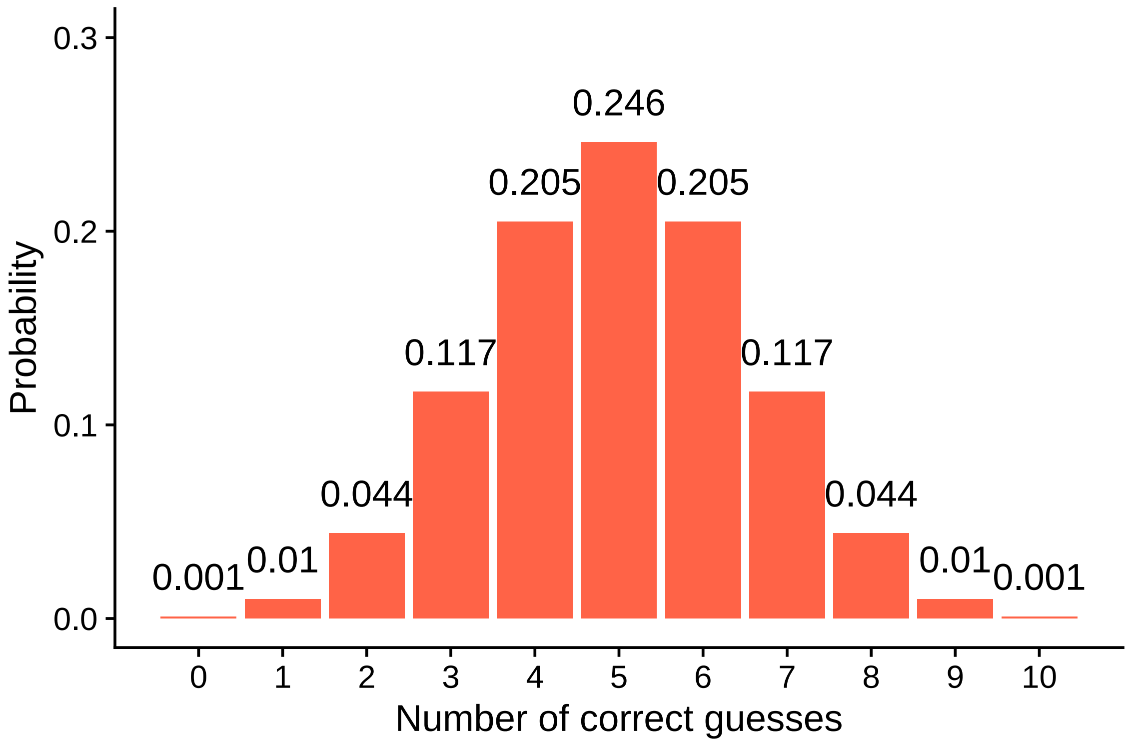
6. What is the most appropriate test statistic for this “study” and what is its numerical value? [1 point]

Answer
The test statistic is the number of correct identifications out of 10 trials: $X = 7$

7. What is the most appropriate null hypothesis? [1 point]

Answer
H_0: I have no real ability to distinguish the two species – I am just guessing at random, so the true probability of correctly identifying a spider is $p = 0.5$

8. The figure below shows the null distribution for the number of correct guesses. Use this figure to calculate the p-value (assume a two-tailed alternative hypothesis). Round your answer to the nearest 0.001. [1 point]



Answer
The observed test statistic is 7 correct guesses. Because the alternative is two-tailed, the p-value is the probability of a result *at least as extreme* as 7 in either direction – that is, 7, 8, 9, or 10 correct, or (by symmetry) 0, 1, 2, or 3 correct:
$P = 0.117 + 0.044 + 0.01 + 0.001 + 0.001 + 0.01 + 0.044 + 0.117 = 0.344$
(equivalently, in R: `2 * sum(dbinom(c(0:3, 7:10), 10, 0.5))`)
$P \approx 0.344$

9. Based on this hypothesis test and a significance level of $\alpha = 0.05$, do you reject the null? What does that mean for how confident I should feel with my spider identification abilities? [1 point]

Answer
Since $P = 0.344 > 0.05$, we **fail to reject the null hypothesis**. Getting 7 out of 10 correct is not at all unusual if you were just guessing at random, so these data do not give us statistical evidence that this "ability" to identify spiders is any better than chance

10. Let's suppose, for my pride (!), that the null hypothesis is not true, has an error occurred in our hypothesis test? If so what kind of error? [1 point]

Answer
If the null hypothesis is actually false (i.e. there really is some true spider-identification skill better than random guessing) but we failed to reject it, that would be a **Type II error** (a false negative)